

Passing Signals at Danger (PSAD)

Revision Guide | GERT8000-S5 | EMR Train Driver Training

1. What Counts as a Stop Signal?

A stop signal is any signal that can display a red aspect or indication. You must not pass a stop signal at danger without authority. Types of stop signal include:

- Main aspect stop signals (colour-light)
- Semaphore stop signals
- Position-light signals associated with a main aspect stop signal
- Intermediate block home signals
- Block markers and limit of shunt signals

Source: S5 s1 / S7

△ CRITICAL: Permission to pass a signal at danger **MUST** always come from the signaller (or another competent person where authorised in the rules). You must never pass a stop signal at danger without authority — except at a closed signal box (see section 9).

2. When a Signal Can Be Passed at Danger

The signaller may authorise you to pass a signal at danger only in the following circumstances:

- The signal is defective or disconnected
- The signal cannot be cleared because signalling or level crossing equipment has failed
- For shunting purposes where there is no position-light or shunting signal
- A movement which has reversed is then required to start from beyond that signal
- An electric train is to pass the signal protecting an isolated section towards the limiting point
- A train accepted on restricted acceptance — line clear only to home signal of next signal box, section signal cannot be cleared
- In an emergency (TCB line only, signal box supervisor or Ops Control authority): passenger train to enter occupied section to use a station platform
- An engineering train is to move towards or leave a possession at an intermediate point
- To pass a signal protecting engineering work to gain access to a station (train required to start back), a line under SLW, or a siding
- To examine the line
- A failed train has been removed from an absolute block section — next train to proceed at caution through that section
- After a train/vehicle has proceeded without authority has been removed, or after front portion of a divided train has passed through
- To enter the section to assist a failed train, evacuate passengers, remove a portion of a divided train, or remove a train that has proceeded without authority
- Single line working applies
- Working by pilot or modified working applies
- An engineering train is to move towards a protection zone

Source: S5 s1.1

3. Getting Authority

Before passing a signal at danger, you must get the personal authority of:

- The signaller, OR
- The shunter acting on the signaller's instructions (shunting movements), OR
- The pilot or handsignaller acting on the signaller's instructions, OR
- Another competent person where authorised in the rules

You must clearly understand:

- Why the signal needs to be passed at danger
- How far the movement can proceed

Source: S5 s1.2

When getting authority: confirm you are speaking to the correct signaller; confirm any intermediate handsignaller is genuine; check you are being authorised for the correct signal; repeat back all safety-critical information.

4. Yellow Handsignals

A yellow handsignal shown at a signal by a handsignaller is authority to pass the signal at danger only if:

- You have stopped your train at the signal and the handsignaller has given you the necessary instructions, OR
- The signaller or pilot has already told you about the circumstances and instructed you to obey the handsignal — in which case you do NOT need to stop if a yellow handsignal is shown at the signal

Source: S5 s3.2

Yellow handsignals with detonators are only used by intermediate handsignallers in connection with SLW in the wrong direction. Be prepared to stop on hearing detonators.

5. Dealing With TPWS

Situation	Action required
Passing a single signal at danger (most circumstances)	Use TPWS train stop OVERRIDE button
Entering a section with Extended Block Working	Use TPWS TEMPORARY ISOLATION switch
Single-line section with pilotman/modified working — passing MORE THAN ONE signal at danger	Use TPWS TEMPORARY ISOLATION switch
Entering a line under possession	Use TPWS TEMPORARY ISOLATION switch

⚠ CRITICAL: You MUST reinstate TPWS before leaving the section when temporary isolation has been used.

Source: S5 s3.4

6. During the Movement: Points and Crossings

If possible, make sure any points, switch diamonds or swing-nose crossings are in the correct position before passing over them.

Circumstance	Maximum speed over points/crossings
Normal PSAD movement	15 mph (25 km/h)
Extended Block Working — after first train has checked points	15 mph initially; up to 50 mph if given permission
Wrong-direction SLW — permission given	Up to 50 mph (80 km/h)

Source: S5 s4.1

7. Speed During the Movement

The default is to proceed at caution throughout, even if the line ahead appears to be clear. Exceptions are:

Maximum speed	Circumstances
Up to 50 mph	SLW in the wrong direction; Modified working on single lines; Extended Block Working; Passing two consecutive main aspect stop signals at danger; Failure of a block instrument on an absolute block line
Up to permissible speed	SLW in the right direction; Single lines where a token is provided and you hold the token; Working by pilot on single or bi-directional lines
At caution	All other PSAD movements (unless told otherwise by signaller)

Source: S5 s4.2

Once you can see that the next stop signal ahead is displaying a proceed aspect, you may travel at any speed appropriate to the aspect — unless the signaller has told you to continue at a reduced speed up to that signal.

Source: S5 s4.4

8. Level Crossings During a PSAD Movement

You must approach at caution and check it is safe before passing over:

- Any controlled level crossing
- Any automatic level crossing that the signaller has told you will not operate normally for the movement
- Any barrow or foot crossing with white-light indications that the signaller has told you will not operate normally

Source: S5 s4.3

9. Ground Position-Light (GPL) Signals

When the signaller authorises you to pass a main aspect stop signal at danger, this allows you to proceed as far as the next main aspect stop signal ahead. You may ignore the aspects of any ground position-light signals or semaphore shunting signals in the route.

However, if the signaller tells you the train is only authorised to proceed up to a specified GPL signal, you must:

- Only proceed as far as that signal
- Stop at it, whatever aspect it is displaying

Source: S5 s4.6

10. Extended Block Working (EBW) — Driver Duties

Extended Block Working is introduced when signalling equipment on a TCB line (not a single line) fails, and consecutive signals must be passed at danger under the same authority.

10.1 At the Entrance Signal

The signaller will tell you that EBW has been introduced and the location of the exit signal. You must:

- Complete an Extended Block Working Ticket (NR3190) at the signaller's dictation
- Tell the guard (if provided) that EBW is in operation

- Operate the TPWS temporary isolation switch before entering the section

Source: S5 s5.5

10.2 During the EBW Movement

If you are the FIRST train into the section, you must:

- Approach all points, switch diamonds and swing-nose crossings at caution
- Check, if possible, that they are in the correct position
- Not pass over any of these at more than 15 mph (25 km/h)

Throughout the EBW section you must not exceed 50 mph.

Proceed at caution if you have entered the section to examine the line, assist a failed train, remove a divided train portion, evacuate passengers, or remove a train that proceeded without authority.

Source: S5 s5.6

10.3 At the Exit Signal

When your train arrives at the exit signal, you must:

- Stop
- IMMEDIATELY reinstate TPWS
- Tell the signaller your train has arrived at the exit signal
- Write "CANCELLED" across the Extended Block Working Ticket
- Obey the signaller's instructions for resuming normal working

Source: S5 s5.7

11. Passing Two Consecutive Main Aspect Stop Signals at Danger

The signaller may authorise you to pass two consecutive main aspect stop signals at danger if ALL of the following conditions are confirmed:

- Because of the same fault, neither signal can display a proceed aspect (or: first at danger, second not displaying any aspect)
- There is no main aspect distant signal between the two signals
- The second main aspect stop signal is NOT fitted with TPWS
- The line has been examined and is clear up to and including the overlap of the next signal beyond the second
- The next signal beyond the second shows a proceed aspect

Source: S5 s6.1

During the movement, you must:

- Pass each of the two main aspect stop signals at danger
- Ignore the aspects of any GPL signals (unless the signaller tells you otherwise)
- Obey the aspects of all subsequent signals
- Travel at no more than 50 mph until you can see the aspect of the next main aspect stop signal beyond the second signal
- Then proceed at a speed appropriate for the aspect displayed by that next signal

Source: S5 s6.4

12. Passing a Signal at a Closed Signal Box

You may only pass a controlled signal at danger on your own authority if you have confirmed the controlling signal box is closed. This applies only on absolute block lines.

Before starting:

- Make sure any points, switch diamonds or swing-nose crossings worked from the closed signal box are set correctly for the movement
- Operate the TPWS train stop override button

During the movement:

- Proceed at caution, even if the line appears to be clear
- Not more than 15 mph over any points, switch diamonds or swing-nose crossings
- Not more than 10 mph through any tunnel
- Pass over any level crossing only when you are sure it is safe to do so

At subsequent signals controlled by the same closed signal box:

- Repeat the above process at each controlled signal at danger

At the next open signal box:

- Contact the signaller there at the first opportunity

Source: S5 s7.1–7.5

13. SPAD — Passing Without Authority

If you pass a signal at danger or pass without authority:

- STOP THE TRAIN IMMEDIATELY
- Tell the signaller that the signal has been passed at danger without authority
- Do not proceed until the signaller gives permission
- Answer all questions the signaller asks you

Source: S5 s8.1

If you see a SPAD indicator illuminated (even on a signal applying to another line):

- Stop the train immediately
- Contact the signaller

Source: S5 s8.2

The signaller may allow the train to be moved to a more convenient location (to clear a junction, level crossing, or bring the train fully into a platform) provided:

- You are prepared to make the movement
- You consider there is no defect that would make it unsafe
- The movement will not proceed beyond another main aspect stop signal or block marker (unless permitted in Signal Box Special Instructions)

⚠ CRITICAL: The train must not proceed until authorised by Operations Control. The signaller must complete form NR3189 and report the incident.

Source: S5 s8.3

Questions & Answers

Q1: Before passing a signal at danger, what must you get, and from whom?

A: The personal authority of the signaller (or shunter acting on signaller's instructions for shunting; or pilot/handsignaller acting on signaller's instructions; or another competent person where authorised in the rules). You must clearly understand why and how far the movement can go.

Source: S5 s1.2

Q2: What is the maximum speed over points and crossings during a normal PSAD movement?

A: 15 mph (25 km/h). This increases to up to 50 mph if you have been given permission during EBW or wrong-direction SLW.

Source: S5 s4.1

Q3: During a PSAD movement, when may you travel at up to 50 mph?

A: During SLW in the wrong direction; during modified working on single lines; during Extended Block Working; when passing two consecutive main aspect stop signals at danger; or during a failure of a block instrument on an absolute block line.

Source: S5 s4.2b

Q4: During a PSAD movement, when may you travel at up to permissible speed?

A: During SLW in the correct direction; on single lines where a token is provided and you hold the token; or during working by pilot on single or bi-directional lines.

Source: S5 s4.2c

Q5: You have been authorised to pass a main aspect signal at danger. Can you ignore the next GPL signal in the route?

A: Yes — authority to pass the main aspect signal at danger allows you to ignore GPL signals in the route and proceed to the next main aspect stop signal ahead. Exception: if the signaller specifically tells you to stop at a named GPL signal, you must stop there regardless of its aspect.

Source: S5 s4.6

Q6: What must you do with TPWS when authorised to pass a single signal at danger (not EBW or SLW)?

A: Use the TPWS train stop OVERRIDE button — not temporary isolation. Temporary isolation is only for EBW, pilotman/modified working on single lines (with more than one signal at danger), and lines under possession.

Source: S5 s3.4

Q7: List the five conditions that must all be met before the signaller can authorise you to pass TWO consecutive main aspect stop signals at danger.

A: (1) Both signals cannot display proceed because of the same fault, or first is at danger and second shows no aspect. (2) No main aspect distant signal between the two. (3) The second signal is NOT fitted with TPWS. (4) The line has been examined and is clear up to and including the overlap of the next signal beyond the second. (5) The next signal beyond the second is displaying a proceed aspect.

Source: S5 s6.1

Q8: You are the first train into an EBW section. What speed applies over points?

A: 15 mph (25 km/h) maximum — you must approach at caution, check the points are in the correct position if possible, and not exceed 15 mph over them.

Source: S5 s5.6

Q9: On arrival at the EBW exit signal, list your immediate actions.

A: Stop. Immediately reinstate TPWS. Tell the signaller your train has arrived at the exit signal. Write "CANCELLED" across the Extended Block Working Ticket. Obey the signaller's instructions for resuming normal working.

Source: S5 s5.7

Q10: You discover you have passed a signal at danger without authority. What must you do

immediately?

A: Stop the train immediately. Tell the signaller that the signal has been passed at danger without authority. Do not proceed until the signaller gives permission. Answer all questions the signaller asks.

Source: S5 s8.1

Q11: You approach a controlled signal at danger and the signal box appears to be closed. What must you confirm before passing it on your own authority?

A: You must confirm that the controlling signal box is closed (this is an absolute block line procedure). Then: check points are set correctly, use TPWS train stop override, proceed at caution, max 15 mph over points, max 10 mph through tunnels, level crossings only when safe.

Source: S5 s7.1–7.3

Q12: During a PSAD movement you reach a level crossing. What must you do?

A: Approach at caution and check it is safe to proceed over any controlled level crossing; any automatic level crossing the signaller has told you will not operate normally; and any barrow or foot crossing with white-light indications the signaller has told you will not operate normally.

Source: S5 s4.3